AI ACCELERATORS: A Long Way To Go

We are witnessing the AI revolution. In order to make smart decisions, the devices equipped with AI need lots and lots of data. And making sense of that data needs power and speed. This is where AI accelerators come into the picture

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As IoT systems are getting more and more efficient, data acquisition is getting easier. This has increased the demand for AI and ML applications in all smart systems. AI based tasks are data-intensive, power-hungry, and call for higher speeds. Therefore, dedicated hardware systems called AI accelerators are used to process AI workloads in a faster and more efficient manner.

Co-processors like graphics processing units (GPUs) and digital signal processors (DSPs) are common in computing systems. Even Intel's 8086 microprocessor can be interfaced with a math co-processor, 8087. These are task-driven additions that were introduced because CPUs alone cannot perform their functions. Similarly, CPUs alone cannot efficiently take care of deep learning and artificial intelligence workloads. Hence, AI accelerators are adopted in such applications. Their designs revolve around multi-core processing, enabling parallel processing that is much faster than the traditional computing systems.

At the heart of ML is the multiply-accumulate (MAC) operation. Deep learning is primarily composed of a large number of such operations. And these need to happen parallelly. AI accelerators have the capacity to significantly reduce the time it takes to perform these MAC operations as well as to train and execute AI models. In fact, Intel's head of architecture, Raja Koduri, noted that one day every chip will be a neural net processor. This year, Intel plans to release its Ponte Vecchio HPC graphics card (accelerator), which, Intel claims, is going to be a game-changer in this arena.

AI in Smartphones

- Apple's iPhone 13 series comes with A15 bionic chips that contain dedicated neural network hardware
- OnePlus 9 Pro and other phones featuring Qualcomm's Snapdragon 888 processor have AI capabilities in their cameras thanks to Qualcomm artificial intelligence (AI) engine
- Samsung Galaxy S20 series includes an AI-powered camera
- Google's Pixel 6's processor has features powered by AI and ML

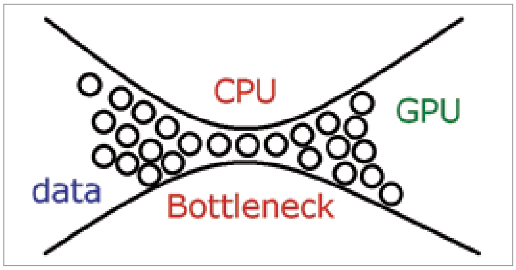

Fig. 1: CPU bottleneck

User-to-hardware expressiveness

'User-to-hardware expressiveness' is a term coined by Adi Fuchs, an AI acceleration architect in a world-leading AI platforms startup. Previously, he has worked at Apple, Mellanox (now NVIDIA), and Philips. According to Fuchs, we are still unable to automatically get the best out of our hardware for a brand new AI model without any manual tweaking of the compiler or software stack. This means we have not yet effectively reached a reasonable user-to-hardware expressiveness.

It is surprising how even slight familiarity with processor architecture and the hardware counterpart of AI can help in improving the performance of our training models. It can help us understand the various bottlenecks that can cause the performance of our models to decrease. By understanding processors, and AI accelerators in particular, we can bridge the gap between writing code and having it implemented on the hardware.

Bottleneck in AI processing

As we add multiple cores and accelerators to our computing engines to boost performance,